



Product Series

INVERTERS

On&Off Grid Inverters



1.2/2.2/4.2KW-12V/24V



6.2/12KW-48V



3.6/6.5/7.2/12KW-48V(120V)

Hybrid Inverters



6/8KW-48V



10/12KW-48V



10/12/15/20KW-48V(3PHASE)

On Grid Inverters



3.6/5KW



36/40/50/60KW



70/80/90/100/110/120KW

ALL IN ONE

All-in-one & Stacked



500/1000W-1000/2000WH



5KW-10/15KWH



4.2KW-8KWH



6.2/12/15KW-10/16/32KWH



7.2/12KW-16KWH(120V)

BATTERY

LiFePO₄



5KWH



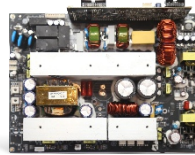
10KWH



16KWH

PCBA

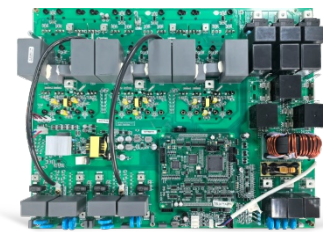
SKD KITS



On & Off Grid Inverter PCBA



Hybrid Inverter PCBA



On Grid Inverter PCBA

INVERTERS

On & Off Grid Inverters

230V Series 1.2-12KW-48V-----7-12

120V Series 3.6-12KW-48V -----13-16

Hybrid Inverters

6-20KW-48V-----17-20

On Grid Inverters

3.6/5KW-----23-24

36-60KW -----25-26

70-120KW-----27-28

230V Series

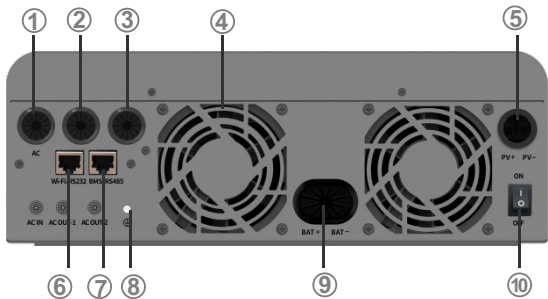


1.2KW-12V 2.2/4.2KW-24V

On & Off Grid Inverters

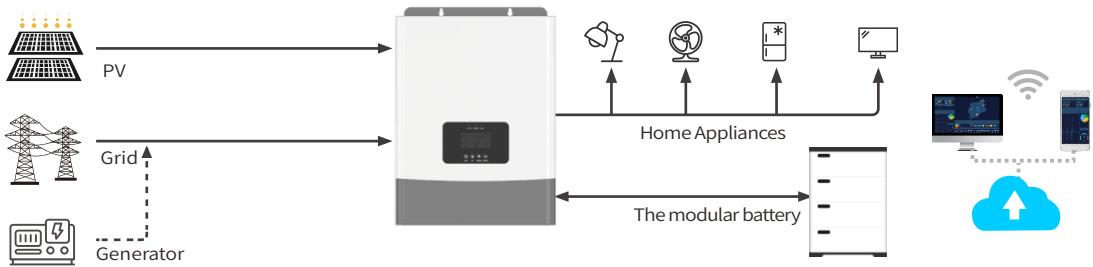
- Features segmented charging and discharging.
- Supports operation without a battery. Solar energy and mains power can supply power to load devices together.
- Equipped with energy-saving mode, reducing no-load losses.
- Offers four charging modes: solar only, mains priority, solar priority, and hybrid charging.
- Provides two output modes: mains bypass and inverter output, and has uninterrupted power supply function.
- Includes complete protection against short-circuit, over-voltage, under-voltage, overload, and backflow, ensuring 360° safety.
- Utilizes fully digital voltage and current dual closed-loop control, advanced SPWM technology, and outputs pure sine waves.
- Enables remote monitoring and management via Wi-Fi or 4G.

Interface Description



- 1. AC/IN connection port
- 2. AC/OUT 1 connection port
- 3. AC/OUT 2 connection port
- 4. Cooling fan
- 5. PV connection port
- 6. Wi-Fi/RS232 communication port
- 7. BMS/RS485 communication port
- 8. Ground terminal
- 9. BAT+ BAT-
- 10. ON/OFF switch

Solar System Connection



Product Specifications

Model	MZ-1.2K-12V-EU	MZ-2.2K-24V-EU	MZ-4.2K-24V-EU
Max. Power (W)	1200VA/1200W	2200VA/2200W	4200VA/4200W
AC Input			
Rated input voltage	230VAC		
Input Voltage	170~280VAC (UPS mode) ; 90~280VAC (APL mode)		
Frequency	50Hz/60Hz (Automatic detection)		
Inverter Output			
Output voltage	230VAC±5%		
Peak power	2000VA	4000VA	8400VA
Efficiency	90%~93%		
Transfer time	≤5ms		
Output voltage waveform	Pure sine wave		
Battery DATA			
Battery voltage	12VDC	24VDC	24VDC
Charging Voltage	13.5VDC	27VDC	27VDC
Battery overvoltage protection value	16VDC	32VDC	32VDC
PV & AC Charging			
Solar charging type	MPPT		
Max. input voltage of MPPT	102VDC	500VDC	
MPPT power	500W	1200W	6000W
MPPT working range	17~80VDC	30~80VDC	120~450VDC
MPPT charging current	40A	120A	
Charging current of mains power	20A	100A	
Max. charging current	60A	120A	
General Data			
EMC certification level	CE-EMC EN IEC 61000-6-1:2019		
Operating temperature range	-20°C to 60°C (Derating 45°C)		
Storage temperature range	-25°C to 55°C		
Humidity range	5% to 95% (Coating protection)		
Noise	≤60dB		
IP Rating	IP21		
Cooling method	Forced air cooling, intelligent variable speed fan		
Communication interface	RS232/RS485		
Dimensions (L * W * H)	338x120x110mm	338x120x110mm	445x326x116mm
Net Weight (kg)	7.0	7.5	7.6

230V Series

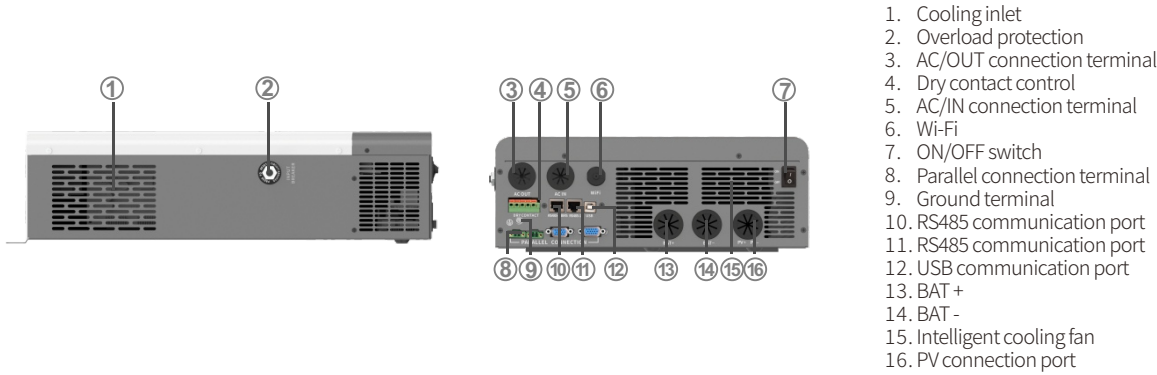


6.2/12KW-48V

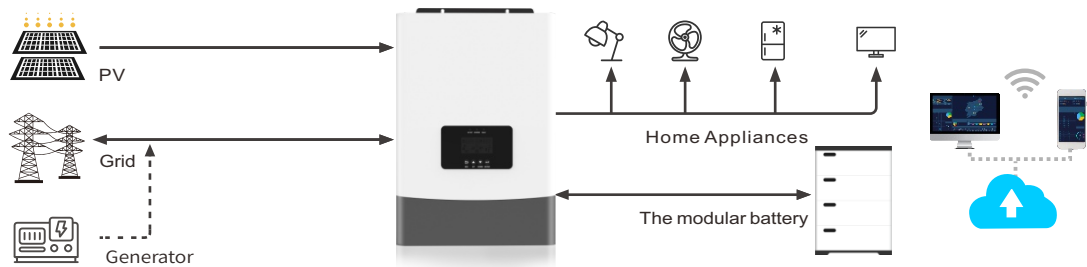
On & Off Grid Inverters

- Features segmented charging and discharging.
- anti-backflow grid connection function (photovoltaic mixed power supply), can set grid connection mode.
- Supports use without battery. Solar energy and mains power can supply power to load devices together.
- Equipped with energy-saving mode functions, reducing no-load losses.
- There are 4 charging modes to choose from: solar only, mains priority, solar priority, and hybrid charging.
- Equipped with dual activation function of lithium battery, it can be triggered by connecting either mains or photovoltaic power source.
- Provides two output modes: mains bypass and inverter output, and has uninterrupted power supply function.
- Includes complete protection against short-circuit, over-voltage, under-voltage, overload, and backflow, ensuring 360° safety.
- Utilizes fully digital voltage and current dual closed-loop control, advanced SPWM technology, and outputting pure sine waves.
- Enables remote monitoring and management via Wi-Fi or 4G.
- Supports PV input up to 18 kW (for the 12 kW model).

Interface Description



Solar System Connection



Product Specifications

Model	MZ-6.2K-48V-EU	MZ-12K-48V-EU
Parallel mode		
Permitted parallel number	1~6	1~6
AC mode		
Rated input voltage	230Vac ±5%	230Vac ±5%
Input voltage range	170~280VAC (UPS mode) ; 90~280VAC (APL mode)	
Frequency	50Hz/60Hz (Auto detection)	
Frequency Range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz);	
Overload/short circuit protection	Circuit breaker	
Efficiency	>93%	
Transfer time	≤5ms	
AC reverse protection	Yes	
Maximum bypass overload current	40A	60A
Inverter mode		
Output voltage waveform	Pure sine wave	
Rated output power (VA/W)	6200VA/6200W	12000VA/12000W
Power factor	1	
Rated output voltage (Vac)	230Vac	
Output voltage error	±5%	
Output frequency range (Hz)	50Hz±0.3Hz/60Hz±0.3Hz	
Maximum Efficiency	>94%	
Overload protection	(102% < load <125%) ±10%: report error and turn off the output after 5 minutes; (125% < load < 150%) ± 10%: report error and turn off the output after 10 seconds; Load >150% ±10%: report error and turn off the output after 5 seconds;	
Peak power	12000VA	24000VA
Loaded motor capability	4HP	8HP
Bypass circuit breaker specification	40A	63A
Rated battery input voltage	51.2V (Minimum starting voltage 44V)	
Battery voltage range	40Vdc~60Vdc±0.6Vdc (Undervoltage alarm/shutdown voltage/overvoltage alarm /overvoltage recovery... settable on LCD screen)	
Power saving mode self-consumption	Load ≤25W	Load ≤50W
AC charging		
Battery type	Lead acid or lithium battery	
Maximum charge current(can be set)	80A	150A
Charge current error	±5Adc	
Charge voltage range	40 ~60Vdc	
Short circuit protection	Circuit breaker and blown fuse	
Circuit breaker specifications	40A	63A
PV charging		
MPPT way	1	2
Maximum PV open circuit voltage	500Vdc	
PV operating voltage range	120Vdc~500Vdc	120Vdc~500Vdc
MPPT voltage range	90Vdc~450Vdc	90Vdc~450Vdc
Battery voltage range	40-60Vdc	
Maximum PV input power	6500W	9000WX2
Maximum PV input current	27A	27A+27A
Charging short circuit protection	Blown fuse	
Wiring protection	Reverse polarity protection	
Hybrid Max. Charging Current(AC+PV)		
Max charging current(can be set)	0-100A	0-150A
General Data		
EMC certification level	CE-EMC EN IEC 61000-6-1:2019	
Operating temperature range	-20°C to 60°C (Derating 45°C)	
Storage temperature range	-25°C to 60°C	
Humidity range	5% to 95% (Conformal coating protection)	
Noise	≤60dB	
IP Rating	IP21	
Cooling method	Forced air cooling, variable speed of fan	
Communication interface	USB/RS485(Wi-Fi/4G)/ Dry contact control	
Wi-Fi function	Yes	
Product Size/Packaging Size (L*W*D)	456x300x125mm/510x370x180mm	579x353x147mm/630x455x215mm
Net weight / Gross weight (kg)	10/11	19/20
Optional function		
Parallel function	☐ YES	☐ NO
Grid anti-backflow CT	☐ YES	☐ NO

230V Series

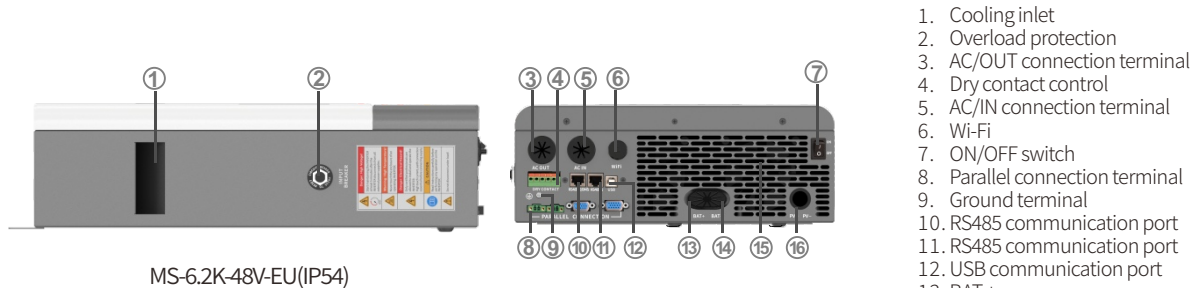


4.2KW-24V 6.2/12KW-48V

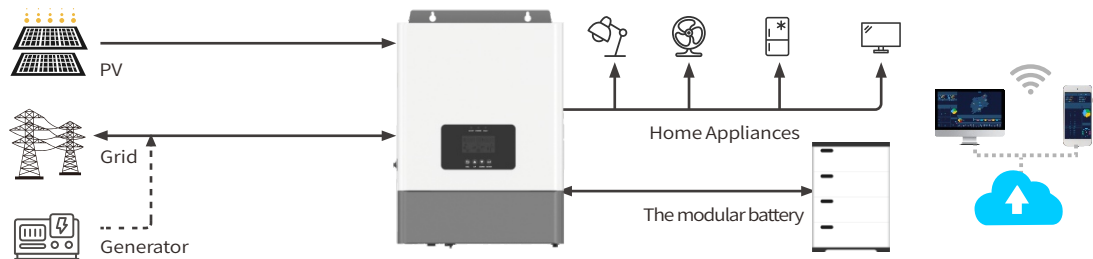
IP54 On & Off Grid Inverters

- Features segmented charging and discharging.
- anti-backflow grid connection function (photovoltaic mixed power supply), can set grid connection mode.
- Supports use without battery. Solar energy and mains power can supply power to load devices together.
- Equipped with energy-saving mode functions, reducing no-load losses.
- There are 4 charging modes to choose from: Solar only, Mains priority, Solar priority, and Hybrid charging.
- Equipped with dual activation function of lithium battery, it can be triggered by connecting either mains or photovoltaic power source.
- Provides two output modes: mains bypass and inverter output, and has uninterrupted power supply function.
- Includes complete protection against short-circuit, over-voltage, under-voltage, overload, and backflow, ensuring 360° safety.
- Utilizes fully digital voltage and current dual closed-loop control, advanced SPWM technology, and outputting pure sine waves.
- Enables remote monitoring and management via Wi-Fi or 4G.
- With IP54 IP Rating

Interface Description



Solar System Connection



Product Specifications

Model	MS-4.2K-24V-EU	MS-6.2K-48V-EU	MS-12K-48V-EU
Parallel Mode			
Permitted parallel number	NO	1~6	1~6
AC Mode			
Rated input voltage	230Vac ±5%		
Input voltage range	170~280VAC (UPS mode) ; 90~280VAC (APL mode)		
Frequency	50Hz/60Hz (Auto detection)		
Frequency Range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz);		
Overload/short circuit protection	Circuit breaker		
Efficiency	>93%		
Transfer time	≤5ms		
AC reverse protection	Yes		
Maximum bypass overload current	30A	40A	60A
Inverter Mode			
Output voltage waveform	Pure sine wave		
Rated output power (VA/W)	4200VA/4200W	6200VA/6200W	12000VA/12000W
Power factor	1		
Rated output voltage (Vac)	230Vac		
Output voltage error	±5%		
Output frequency range (Hz)	50Hz±0.3Hz/60Hz±0.3Hz		
Maximum Efficiency	>94%		
Overload protection	(102% < load <125%) ±10%: report error and turn off the output after 5 minutes; (125% < load < 150%) ± 10%: report error and turn off the output after 10 seconds; Load >150% ±10%: report error and turn off the output after 5 seconds;		
Peak power	8400VA	12000VA	24000VA
Loaded motor capability	2HP	4HP	8HP
Bypass circuit breaker specification	32A	40A	63A
Rated battery input voltage	24V	51.2V (Minimum starting voltage 44V)	
Battery voltage range	20.0~30Vdc±0.6Vdc	40.0Vdc~60Vdc±0.6Vdc (Undervoltage alarm/shutdown voltage/overvoltage alarm/overvoltage recovery... settable on LCD screen)	
Power saving mode self-consumption		Load≤25W	Load≤50W
AC Charging			
Battery type	Lead acid or lithium battery		
Maximum charge current(can be set)	80A	80A	150A
Charge current error		±5Adc	
Charge voltage range	28~30Vdc	40 ~60Vdc	
Short circuit protection		Circuit breaker and blown fuse	
Circuit breaker specifications	40A	40A	63A
PV Charging			
MPPT way	1	1	2
Maximum PV open circuit voltage	500VDC	500Vdc	
PV operating voltage range	120~450VDC	120Vdc~500Vdc	120Vdc~500Vdc
MPPT voltage range	90Vdc~450Vdc	90Vdc~450Vdc	90Vdc~450Vdc
Battery voltage range		40~60Vdc	
Maximum PV input power	6000W	6500W	9000WX2
Maximum PV input current	20A	27A	27A+27A
Charging short circuit protection		Blown fuse	
Wiring protection		Reverse polarity protection	
Hybrid Max. Charging Current(AC+PV)			
Max charging current(can be set)	0~100A	0~100A	0~150A
General Data			
Operating temperature range	-20°C to 60°C (Derating 45°C)		
Storage temperature range	-25°C to 60°C		
Humidity range	5% to 95% (Conformal coating protection)		
Noise	≤60dB		
IP Rating	IP54		
Cooling method	Forced air cooling, variable speed of fan		
Communication interface	USB/RS485(Wi-Fi/4G)/ Dry contact control		
Wi-Fi function	NO	YES	
Product Size/Packaging Size (L*W*D)	450x362x125mm/510x440x205mm	486.5x325x125mm/550x375x200mm	610x458x150mm/665x530x220mm
Net weight / Gross weight (kg)	9/10.5	10/11	19.5/21
Optional Function			
Parallel function	NO	☐ YES	☐ NO
Grid anti-backflow CT	NO	☐ YES	☐ NO

120V Series

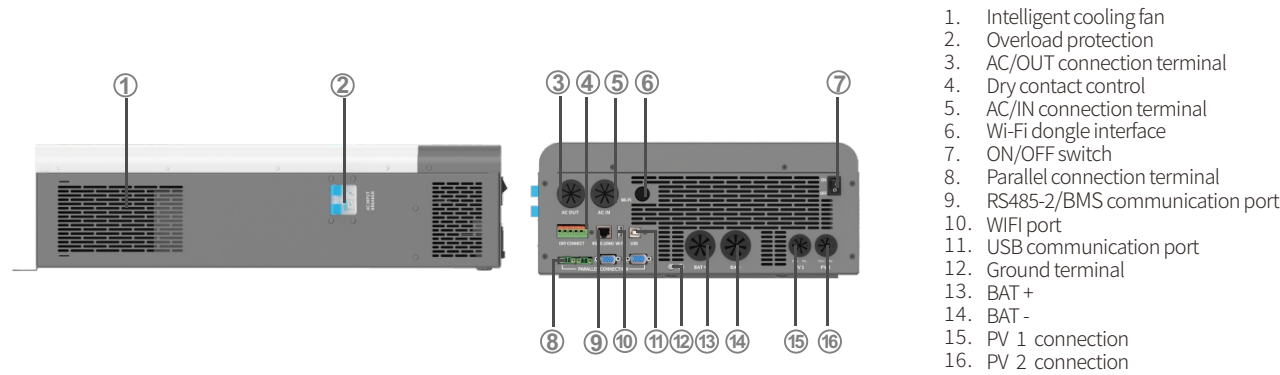


3.6/6.5KW-48V

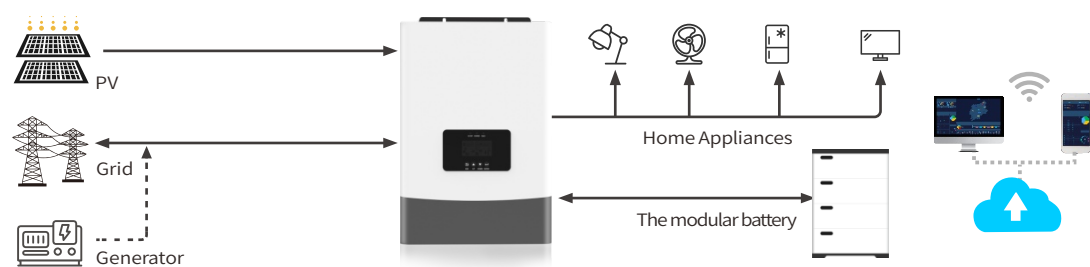
On & Off Grid Inverter

- Features segmented charging and discharging.
- anti-backflow grid connection function (photovoltaic mixed power supply), can set grid connection mode.
- Supports use without battery. Solar energy and mains power can supply power to load devices together.
- Equipped with energy-saving mode functions, reducing no-load losses.
- There are 4 charging modes to choose from: solar only, mains priority, solar priority, and hybrid charging.
- Equipped with dual activation function of lithium battery, it can be triggered by connecting either mains or photovoltaic power source.
- Provides two output modes: mains bypass and inverter output, and has uninterrupted power supply function.
- Includes complete protection against short-circuit, over-voltage, under-voltage, overload, and backflow, ensuring 360° safety.
- Utilizes fully digital voltage and current dual closed-loop control, advanced SPWM technology, and outputting pure sine waves.
- Enables remote monitoring and management via Wi-Fi or 4G.

Interface Description



Solar System Connection



Product Specifications

Model	SH-3.6K-48V-US	SH-6.5K-48V-US
Parallel Mode		
Permitted parallel number	1~6	1~6
AC Mode		
Rated input voltage	120Vac ±5%	
Input voltage range	(90Vac~140Vac)±2%	
Frequency	50Hz/60Hz (Auto detection)	
Frequency Range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz);	
Overload/short circuit protection	Circuit breaker	
Efficiency	>95%	
Transfer time	≤5ms	
AC reverse protection	Yes	
Maximum bypass overload current	40A	60A
Inverter Mode		
Output voltage waveform	Pure sine wave	
Rated output power (VA/W)	3600VA/3600W	6500VA/6500W
Power factor	1	
Rated output voltage (Vac)	120Vac	
Output voltage error	±5%	
Output frequency range (Hz)	50Hz±0.3Hz/60Hz±0.3Hz	
Maximum Efficiency	>91%	
Overload protection	(102% < load <110%) ±10%: report error and turn off the output after 5 minutes; (110% < load < 125%) ± 10%: report error and turn off the output after 10 seconds; Load >125% ±10%: report error and turn off the output after 5 seconds;	
Peak power	7200VA	12000VA
Loaded motor capability	2HP	5HP
Output short circuit protection	Circuit breaker	
Bypass circuit breaker specification	40A	63A
Rated battery input voltage	48V (Minimum starting voltage 44V)	
Battery voltage range	40.0Vdc~60Vdc±0.6Vdc (Undervoltage alarm/shutdown voltage/overvoltage alarm /overvoltage recovery... settable on LCD screen)	
Power saving mode self-consumption	Load≤25W	
AC Charging		
Battery type	Lead acid or lithium battery	
Maximum charge current(can be set)	40A	120A
Charge current error	±5Adc	
Charge voltage range	40 ~60Vdc	
Short circuit protection	Circuit breaker and blown fuse	
Circuit breaker specifications	40A	63A
PV Charging		
MPPT channels	1	2
Maximum PV open circuit voltage	300Vdc	
PV operating voltage range	120Vdc~300Vdc	
MPPT voltage range	90Vdc~260Vdc	
Battery voltage range	40-60Vdc	
Maximum PV input power	4500W	4500W+4500W
Maximum PV input current	27A	22A+22A
Wiring protection	Reverse polarity protection	
Hybrid Max. Charging Current(AC+PV)		
Max charging current(can be set)	0-80A	0-150A
General Data		
Operating temperature range	-20°C to 60°C (Derating 45°C)	
Storage temperature range	-25°C to 60°C	
Humidity range	5% to 95% (Conformal coating protection)	
Noise	≤60dB	
IP Rating	IP21	
Cooling method	Forced air cooling, variable speed of fan	
Communication interface	USB/RS485(Wi-Fi/4G)/ Dry contact control	
Size (L*W*D)	415x280x100mm	550x365x130mm
Net Weight (kg)	7.8	21
Optional Function		
Parallel function	☐ YES ☐ NO	
Wi-Fi function	☐ YES ☐ NO	
Grid anti-backflow CT	☐ YES ☐ NO	

120V Series

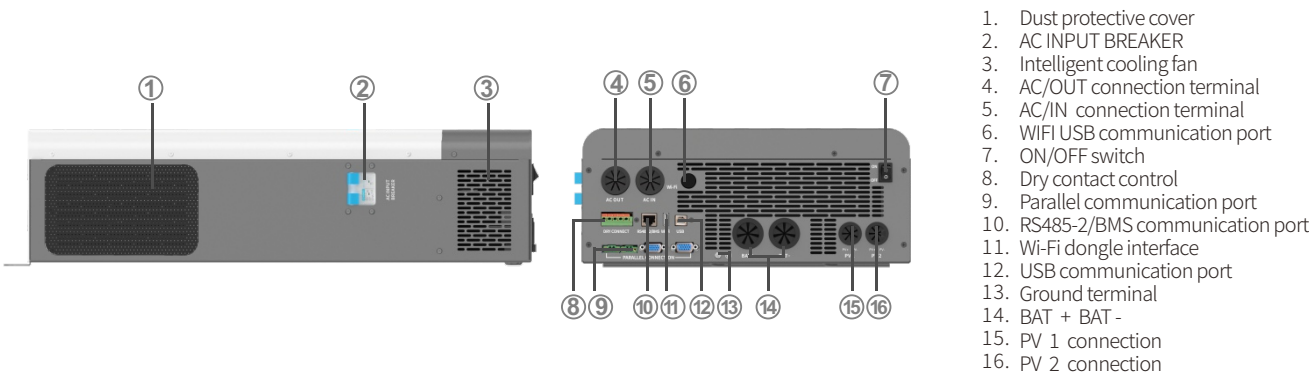


7.2/12KW-48V

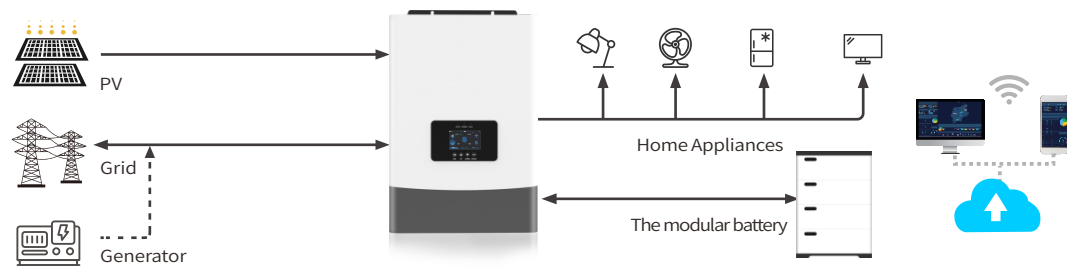
Split Phase On & Off Grid Inverter

- Features segmented charging and discharging.
- anti-backflow grid connection function (photovoltaic mixed power supply), can set grid connection mode.
- Supports use without battery. Solar energy and mains power can supply power to load devices together.
- Equipped with energy-saving mode functions, reducing no-load losses.
- There are 4 charging modes to choose from: solar only, mains priority, solar priority, and hybrid charging.
- Equipped with dual activation function of lithium battery, it can be triggered by connecting either mains or photovoltaic power source.
- Provides two output modes: mains bypass and inverter output, and has uninterrupted power supply function.
- Includes complete protection against short-circuit, over-voltage, under-voltage, overload, and backflow, ensuring 360° safety.
- Utilizes fully digital voltage and current dual closed-loop control, advanced SPWM technology, and outputting pure sine waves. supports split-phase and single-phase pure sine wave output.
- Diverse human-computer interaction: support CAN, USB, Wi-Fi and RS485 communication, local and remote power on or off, Dry contact control.

Interface Description



Solar System Connection



Product Specifications

Model	SH-7.2K-48V-US		SH-12K-48V-US	
Parallel Mode				
Permitted parallel number	1~6		1~6	
AC Mode				
Rated input voltage	120(L1/N, L2/N)/240Vac(L1/L2)			
Input phase voltage range	(85Vac~140Vac)±2%			
Frequency	50Hz/60Hz (Automatic detection)			
Frequency Range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz);			
Overload/short circuit protection	Circuit breaker			
Efficiency	>95%			
Transfer time	≤5ms			
AC reverse protection	Yes			
Maximum bypass phase overload current	2 pole, 63A/63A(L1/L2)			
Inverter Mode				
Output voltage waveform	Pure sine wave			
Rated output power (VA)	Split phase: 7200 Single phase: 6000 (limited by the wiring terminals)		Split phase: 12000 Single phase: 7200 (limited by the wiring terminals)	
Rated output power (W)	Split phase: 7200 Single phase: 6000		Split phase: 12000 Single phase: 7200 (limited by the wiring terminals)	
Power factor	1			
Rated output voltage (Vac)	120Vac (L1/N, L2/N)/240Vac(L1/L2)			
Output voltage error	±5%			
Output frequency range (Hz)	50Hz±0.3Hz/60Hz±0.3Hz			
Maximum Efficiency	>91%			
Overload protection	(102% < load <110%) ±5%: report error and turn off the output after 5 minutes; (110% < load < 125%) ± 5%: report error and turn off the output after 10 seconds; Load >125% ±5%: report error and turn off the output after 5 seconds;			
Peak power	Split phase: 10800VA Single phase: 10800VA		Split phase: 18000VA Single phase: 18000VA	
Loaded motor capability	4HP		6HP	
Output short circuit protection	Can protect, after continuous short circuit for 1 second			
Bypass circuit breaker phase specification	2 pole, 63A/63A(L1/L2)			
Rated battery input voltage	51.2V (Minimum starting voltage 44V)			
Battery voltage range	40.0Vdc~58.6Vdc±0.6Vdc (Undervoltage alarm/shutdown voltage/overvoltage alarm /overvoltage recovery... settable on LCD screen)			
Power saving mode self-consumption	Load≤25W enter power-saving mode			
AC Charging				
Battery type	Lead acid or lithium battery			
Maximum charge current(can be set)	120A		160A	
Charge current error	±5Adc			
Charge voltage range	40 ~58.6Vdc			
Short circuit protection	Circuit breaker and blown fuse			
Circuit breaker phase specifications	2 pole, 63A/63A(L1/L2)			
PV Charging				
Number of MPPT groups	2			
Maximum PV open circuit voltage	500Vdc			
PV operating voltage range	90Vdc~485Vdc		90Vdc~500Vdc	
MPPT voltage range	90Vdc~450Vdc			
Battery voltage range	40-58.6Vdc			
Maximum PV input power	4500W+4500W		9000W+9000W	
Maximum PV input current	16A+16A		27A+27A	
Charging short circuit protection	Blown fuse			
Hybrid Max. Charging Current(AC+PV)				
Max charger current(can be set)	0-150A		0-220A	
General Data				
Operating temperature range	-20°C to 60°C (Derating 45°C)			
Storage temperature range	-25°C to 60°C			
Humidity range	5% to 95% (Conformal coating protection)			
Noise	≤60dB			
Cooling method	Forced air cooling, variable speed of fan			
Communication interface	USB/RS485(Wi-Fi/4G)/ Dry contact control			
IP Rating	IP21			
Product /Packaging dimensions (L*W*H)	599x370x136mm/700x485x212mm		620x450x132mm/758x564x224mm	
Net weight / Gross weight (kg)	20/21.5		23.8/28.6	
Optional Function				
Parallel function	☐ YES		☐ NO	
Wi-Fi function	☐ YES		☐ NO	
Grid anti-backflow CT	☐ YES		☐ NO	

230V Series

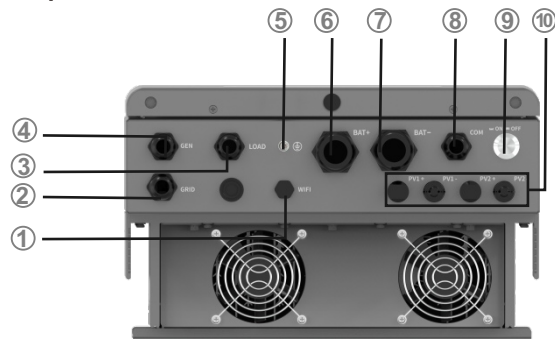


6/8KW-48V

IP66 Hybrid Inverter

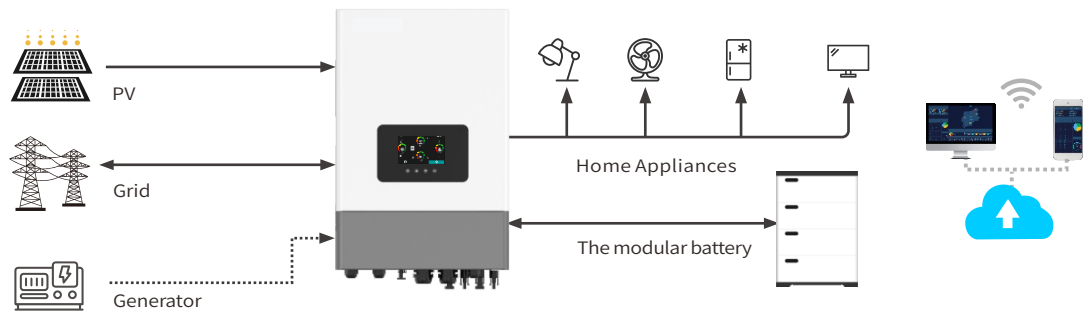
- Advanced MPPT with up to 99.9% efficiency
- Up to 22A+22A PV input current
- Comprehensive security from hardware to software
- Multiple safety approvals
- Up to 100A charging current
- Supports Li-ion battery BMS communication
- Outputs high-quality pure sine wave AC power
- With IP66 IP Rating
- Easy to install and simple to use
- Exclusive Li-ion battery BMS dual activation
- Time-slot function to save cost with peak-valley
- PV input up to 12 kW.

Interface Description



- 1. Wi-Fi dongle interface
- 2. Grid interface
- 3. Load port
- 4. Generator
- 5. Grounding terminal
- 6. BAT +
- 7. BAT -
- 8. Communication port
- 9. ON/OFF Switch
- 10. PV 1/ PV 2

Solar System Connection



Product Specifications

Model	ME-6K-48V-EU		ME-8K-48V-EU	
OFF-Grid Output				
Rated Output Power	6000W		8000W	
Max. Peak Power	12000VA		16000VA	
Rated Output Voltage	230Vac, single phase			
Load Capacity of Motors	4HP			
Output Frequency Range	50Hz ± 0.3Hz/60Hz ± 0.3Hz			
Transfer time	≤5ms			
ON-Grid Output				
Rated Output Power	6000W		8000W	
Max. Apparent Power	6000VA		8000VA	
Rated. Output Current	26A		34.8A	
Max. Output Current	28.6A		38.3A	
THD	< 3%			
Power Factor	0.8 leading to 0.8 lagging			
Rated Voltage	230Vac, single phase			
Frequency	50Hz/60Hz			
Battery				
Battery Type	Lead-acid / Li-ion / User-defined			
Rated Battery Voltage	48V (minimum start-up voltage 44V)			
Battery Voltage Range	40~60Vdc			
Max. PV Charging Current	100A			
Max. Utility Charging Current	60A		80A	
Max. Generator Charging Current	40A		60A	
PV Input				
Number of MPPT	2			
Max. Input Power	4500W+4500W		6000W+6000W	
Max. Input Current	18A+18A		22A+22A	
Max. Short-circuit Current	27A+27A			
Max. Open-circuit Voltage	500Vdc+500Vdc			
MPPT Operating Voltage Range	120~450Vdc			
Grid / Generator Input				
Input Voltage Range	170~280Vac			
Input Frequency Range	50/60Hz			
Bypass Overload Current	40A			
Efficiency				
MPPT Tracking Efficiency	99.9%			
Max. Efficiency	97.5%			
General Data				
Number of Parallel	1~6 units			
Product dimensions (L*W*H)	556*345*205mm			
Packaging dimensions (L*W*H)	635*420*285mm			
Net weight (kg)	20kg		21kg	
Gross weight (kg)	21.5kg		25.2kg	
IP Rating	IP66			
Operating Temperature Range	-20°C to 60°C (Derating 45°C)			
Humidity Range	0% ~ 100%			
Noise	<60dB			
Cooling method	heatsink & intelligent air cooling			
Warranty	5 years			
Communication				
Internal Interface	Rs485 / CAN / USB / Dry contact control			
External Module (optional)	Wi-Fi / 4G			
EMC certification level	CE-EMC EN IEC 61000-6-1:2019			
RoHS	Yes			
Wi-Fi function	Yes			
Optional Function				
Parallel function	☐ YES ☐ NO			
Grid anti-backflow CT	☐ YES ☐ NO			

230V Series

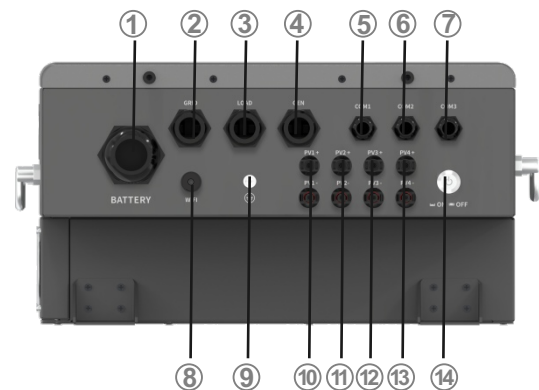


10/12KW-48V

IP66 Hybrid Inverter

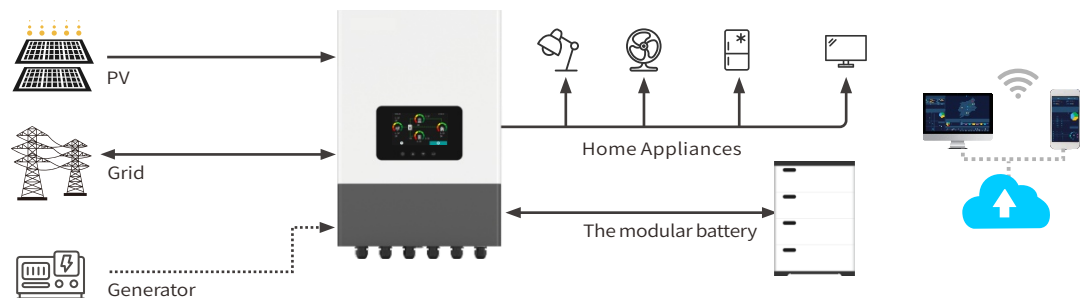
- Advanced MPPT with up to 99.9% efficiency
- Up to 27A X 2 PV input current
- With software and hardware security protection
- Multiple safety approvals
- Up to 200A charging current
- Supports Li-ion battery BMS communication
- Outputs high-quality pure sine wave AC power
- With IP66 IP Rating
- Easy to install and simple to use
- Exclusive Li-ion battery BMS dual activation
- Time-slot function to save cost with peak-valley

Interface Description



- 1. Battery
- 2. Grid interface
- 3. Load port
- 4. Generator
- 5. Communication port 1
- 6. Communication port 2
- 7. Communication port 3
- 8. Wi-Fi dongle interface
- 9. Ground terminal
- 10. PV 1
- 11. PV 2
- 12. PV 3
- 13. PV 4
- 14. ON/OFF Switch

Solar System Connection



Product Specifications

Model	ME-10K-48V-EU		ME-12K-48V-EU	
Inverter Output				
Rated Output Power	10000W		12000W	
Max. Peak Power	20000VA		24000VA	
Rated Output Voltage	230Vac single-phase / three-phase (parallel)			
Load Capacity of Motors	6HP			
Output Frequency	50/60Hz			
Number of Parallel	1~6 units			
AC Output (Grid)				
Rated Output Power	10000W		12000W	
Max. Apparent Power	10000VA		12000VA	
Max. Output Current	43.5A		52.2A	
Max. Output Current	47.8A		57.4A	
Rated Output Voltage	230Vac			
THD	< 3%			
Power Factor	0.8 leading to 0.8 lagging			
Rated Voltage	230Vac			
Frequency				
Battery				
Battery Type	Lead-acid / Li-ion / User-defined			
Rated Battery Voltage	51.2V			
Battery Voltage Range	40~60Vdc			
Max. PV Charging Current	200A			
Max. Utility Charging Current	120A			
Max. Generator Charging Current	60A			
PV Input				
Number of MPPT	2/2			
Max. Input Power	7500W+7500W		9000W+9000W	
Max. Input Current	27A+27A			
Max.Voltage of Open Circuit	500Vdc+500Vdc			
MPPT Voltage Range	120-450Vdc			
Mains/Generator Input				
Input Voltage Range	130~280Vac			
Frequency Range	50/60Hz			
Bypass Overload Current	63A			
Efficiency				
MPPT Tracking Efficiency	99.9%			
Max. Efficiency	97.5%			
European Efficiency	97%			
General Data				
Product dimensions (L*W*H)	750*500*246mm			
Packaging dimensions (L*W*H)	800*550*330mm			
Net weight (kg)	52kg			
Gross weight (kg)	54kg			
IP Rating	IP66			
Operating Temperature Range	-20°C to 60°C (Derating 45°C)			
Humidity range	0~100%			
Cooling method	Forced air cooling			
Communication				
Embedded Interfaces	Rs485 / CAN / USB / Dry contact control			
External Module (optional)	Wi-Fi / 4G			
EMC certification level	CE-EMC EN IEC 61000-6-1:2019			
RoHS	Yes			
Wi-Fi function	Yes			
Optional Function				
Parallel function	☐ YES		☐ NO	
Grid anti-backflow CT	☐ YES		☐ NO	

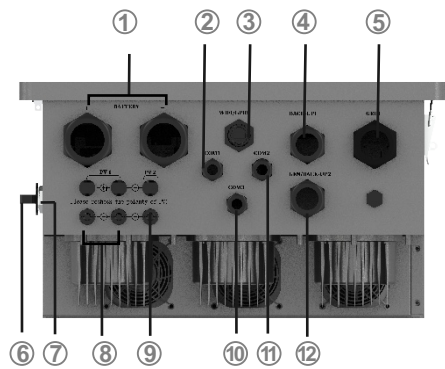
230V Series



10/12/15/20KW-48V
IP66 Three-Phase Hybrid Inverter

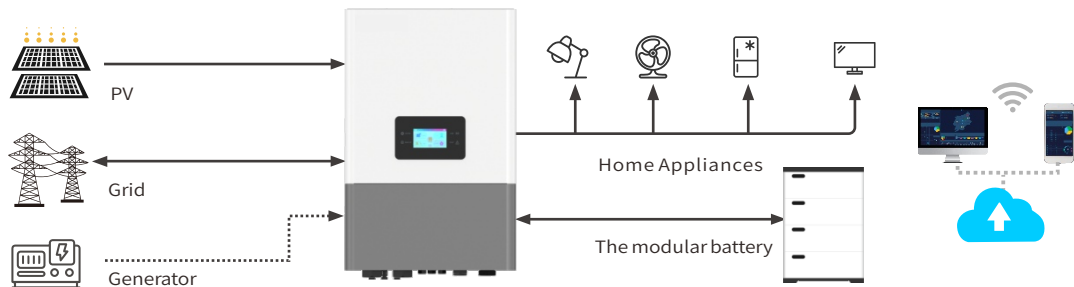
- For Grid-tied and Off-Grid Operation Supports Multiple Batteries in Parallel.
- 100% Unbalanced Output.
- Supports Storing Energy from Diesel Generators.
- Maximum Charging/Discharging Current: 250A

Interface Description



- 1. Battery
- 2. Communication port 1
- 3. Wi-Fi /GPRS dongle interface
- 4. BACK-UP1
- 5. Grid interface
- 6. PV Switch
- 7. DC Switch(Optional)
- 8. PV 1
- 9. PV 2
- 10. Communication port 3
- 11. Communication port 2
- 12. GEN/BACK-UP2

Solar System Connection



Product Specifications

Model	AT-10K-48V-EU	AT-12K-48V-EU	AT-15K-48V-EU	AT-20K-48V-EU
Battery Input Data				
Battery Type	Lithium or Lead-acid Battery			
Rated Battery Voltage (V)	48V			
Max. Charging Voltage (V)	≤60V(configurable)			
Max. Charging/Discharging Current (A)	210A	250A	280A	350A
PV Input Data				
Max. DC Input Power(KW)	20KW	24KW	30KW	32KW
Max. DC Input Voltage(V)	1000			
MPPT Voltage Range (V)	200~800V			
Start-up Voltage (V)	150V			
Max. Input Current(A)	36A/18A			36A/36A
Number of MPPT Trackers	2			
AC Output Data(Grid-tied)				
Max. Apparent Output Power(VA)	11000	13200	16500	22000
Rated Output Voltage(V)	380/400V			380/400V 3W+N+PE
Rated Output Frequency (Hz)	50/60Hz			
Maximum Output Current (A)	16.7A	20A	25A	32A
Output Power Factor	1(0.8 leading~0.8lagging)			
AC Output Data (Off-grid)				
Rated Output Power(W)	10000	12000	15000	20000
Peak Output Power(W)	>200%,15sec			
Rated Output Voltage (V)	380/400V			230/400V 3W+N+PE
Rated Output Frequency (Hz)	50/60Hz			
Rated Output Current(A)	16.7A	20A	24A	32A
Efficiency				
Max.Efficiency(PV)	98%			
Max.Efficiency(Battery)	94.5%			98.5%
European Efficiency	97.5%			98.2%
Protection				
PV Perverse Polarity Protection	Integrated			
PV Insulation Resistance Detection	Integrated			
Residual Current Detection	Integrated			
Output Overcurrent Protection	Integrated			
Output Short-circuit Protection	Integrated			
Output Overvoltage Protection	Integrated			
General Data				
Operating Temperature(°C)	-20°C to 60°C (Derating 45°C)			
Storage Temperature(°C)	-30 to 65°C			
Relative Humidity	0~95%			
Operating Altitude (m)	≤4000m(derating above 2000m)			
Cooling method	Intelligent Forced Air Cooling			
Noise(dB)	<55dB			
Weight(kg)	38KG			43KG
Dimensions(WxHxD mm)	475X683X256mm			750X475X256mm
IP Rating	IP66			
Topology	High-frequency Isolation(Battery Side)			
Safety/EMC Certification	IEC62109,ICE61000,IS16169&IS16221(BIS)			
Grid Compliance	IEC61727,EN50549-1,VDE-4105,NRS-097-2-1,OVE-Richtlinie R25,UNE217001/2,Ordinance No.140			
Other Certifications	IEC61683, IEC62116, EN50530,IEC60068			
Optional Function				
Parallel function	☐ YES		☐ NO	
Wi-Fi function	☐ YES		☐ NO	
Grid anti-backflow CT	☐ YES		☐ NO	

On-Grid Series

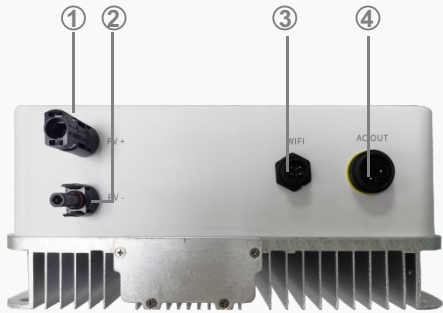
3.6/5KW

IP66 On-Grid Inverter



- 150% DC Input Oversizing
- Wide MPPT voltage range: 50V-550V
- Max. input current per string: 20A, Compatible with high power modules
- Smart I-V Curve Diagnosis Function: locate PV string faults accurately and automatically detect faults
- Intelligent Fault Detection: AC side voltage and current waveforms real-time recorded, fast fault location
- Support RS485/WiFi/4G: remote monitoring and operation via PC or mobile phones
- IP66 Protection Degree: support outdoor installation
- DC & AC Type II SPD: prevent lightning damage
- AFCI Function (Optional): when an arc-fault is detected the inverter immediately stops operation

Interface Description



- 1. PV connection +
- 2. PV connection -
- 3. WIFI stick interface
- 4. AC OUT connection terminal

Solar System Connection



Product Specifications

Model	YG-3.6K-M1-EU	YG-5K-M1-EU
Input(DC))		
Max. Input Power	5.52 KW	7.5KW
Max. Input Voltage	600V	
Start Voltage	50V	
Rated Input Voltage	360V	
MPPT Voltage Range	50-550V	
Number of MPPT Trackers / String per MPPT	1/1	
Max. Current per MPPT	20A	
Max. Short Circuit Current per MPPT	26A	
Output(AC)		
Max. Output Current	16A	22.7A
Rated Output Power	3.68KW	5KW
Max. Output Power	3.68KVA	5KVA
Rated Grid Frequency	50/60Hz	
Rated Grid Voltage	220Vac/230Vac/240Vac	
Power Factor	0.8 lead... 0.8 lag	
THDi	<3%(Rated power)	
Efficiency		
Max. Efficiency	97.8%	
European Efficiency	97.3%	
MPPT Efficiency	99.9%	
Protection		
DC Reverse Polarity Protection	Yes	
Anti-islanding Protection	Yes	
AC Short Circuit Protection	Yes	
Residual Current Monitoring Unit	Yes	
Insulation Resistance Monitoring	Yes	
Ground Fault Monitoring	Yes	
Grid Monitoring	Yes	
PV String Monitoring	Yes	
Surge Protection	Type II	
AFCI Protection	Optional	
General Data		
Dimensions(L*W*H)	270x250x145 mm	
Net Weight(kg)	13	
Operating Temperature Range	-15°C to 55°C	
Cooling Method	Natural	
Protection Degree	IP66	
Max. Operating Altitude	4000m	
Relative Humidity	0~100%	
Topology	Transformerless	
Night Power Consumption	<1 w	
Communication	RS485/USB,4G/WIFI (Optional)	
Standard Compliance		
Certification	G98/G99,VDE0126,VDE0124, EN50549-1/2,CEI 0-21/CEI 0-16,AS4777.2,IEC62116	
Optional function		
Parallel function	☐ YES	☐ NO
WIFI function	☐ YES	☐ NO
Grid anti backflow CT	☐ YES	☐ NO

On-Grid Series



36/40/50/60KW

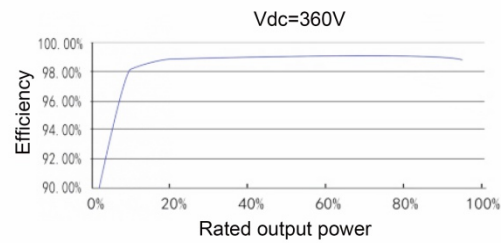
IP66 On-Grid Inverter

- **Flexible Design**
Multiple interfaces: RS485, optional Wi-Fi/4G
Built-in DC breaker for safety & maintenance
Integrated combiner box & DC lightning protection, reducing system cost.
- **Efficient Conversion**
Transformerless, max. efficiency up to 99.02% (European efficiency up to 98.51%)
THD < 3%
3-level SVPWM control for high DC voltage utilization
- **Grid-Friendly**
LVRT/HVRT compliant
Adjustable Power Factor (0.8 leading to 0.8 lagging)
Active and passive anti-islanding protection
0-100% continuously adjustable active power.
- **Excellent qualities**
CQC Gold Sun Certification, TUV Certification, SAA Certification, CE Certification

Solar System Connection



Efficiency Curve



Product Specifications

Model	AT-36K-TLC-EU	AT-40K-TLC-EU	AT-50K-TLC-EU	AT-60K-TLC-EU
Input Data				
Max.DC input power	54000W	60000W	75000W	90000W
Max.DC input voltage	1000V			
Max.DC input current	40/40/20/40/40/20A		40/40/20/20/40/40/20/20A	
MPPT voltage range	180-1000V			
Recommended MPPT operating voltage	360V			
Starting voltage	180V			
Number Of MPPT	6		8	
Max. Number of strings per MPPT	2			
Output Data				
Rated output power	36000W	40000W	50000W	60000W
Max.output power	39.6kVA	44kVA	55kVA	66kVA
Max.output current	103.9A	115.4A	144.3A	173.2A
Rated grid voltage	3/(N)/PE, 220V			
Grid voltage range	170-265Vac			
Rated grid frequency	50Hz/60Hz			
Grid frequency range	47.5-52.5Hz/57.5-62.5Hz			
THD	<3% (Under rated power)			
Power factor	>0.99 (Under rated power) / Adjustable range: 0.8 leading-0.8 lagging			
DC current injection	<0.5% (Under rated power)			
System Data				
Max. efficiency	98.70%	98.70%	98.80%	98.80%
European efficiency	98.10%	98.20%	98.20%	98.30%
Humidity range	0-100% non-condensing			
Cooling method	Intelligent forced air cooling			
Temperature range	-20°C to 60°C (Derating 45°C)			
Night-time power consumption	4000m			
Max. working altitude	LED(optional: LCD)			
Display	Wi-Fi(optional: RS485 or 4G)			
Communication interface				
Protection				
DC reverse-polarity protection	YES			
Short circuit protection	YES			
Output overcurrent protection	YES			
Output overvoltage protection	YES			
Insulation resistance monitoring	YES			
Residual current detection	YES			
Surge protection	YES			
Grid monitoring	YES			
Islanding protection	YES			
Temperature protection	YES			
Integrated DC switch	YES			
Mechanical Data				
Dimensions (W*H*D)	1160*611*286mm			
Weight	90kg			
IP Rating	IP66			
Standard				
Safety standard/EMC standard	NB/T32004-2018, IEC62109, IEC61000, IS16169 & IS16221(BIS)			
Grid-connected standard	IEC61727, EN50549-1, VDE-4105, NRS-097-2-1, OVE-Richtlinie R25, UNE217001/2, Ordinance No.140			
Other standard	Rated output power IEC61683, IEC62116, EN50530, IEC60068			
Optional Function				
Wi-Fi function	☐ YES ☐ NO			
Grid anti-backflow CT	☐ YES ☐ NO			

On-Grid Series



70/80/90/100/110/120KW

IP66 On-Grid Inverter

- Flexible Design**

Multiple interfaces: RS485, optional Wi-Fi/4G

Built-in DC breaker for safety & maintenance

Integrated combiner box & DC lightning protection, reducing system cost.
- Efficient Conversion**

Transformerless, max. efficiency up to 99% (European efficiency up to 98.5%)

THD < 2%

3-level SVPWM control for high DC voltage utilization
- Grid-Friendly**

LVRT/HVRT compliant

Adjustable Power Factor (0.8 leading to 0.8 lagging)

Active and passive anti-islanding protection

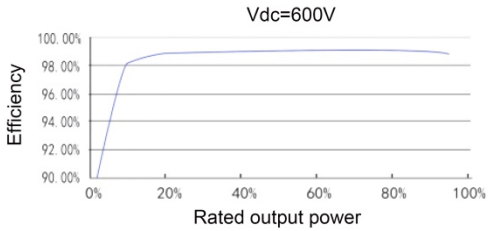
0-100% continuously adjustable active power
- Excellent qualities**

CQC Gold Sun Certification, TUV Certification, SAA Certification, CE Certification

Solar System Connection



Efficiency Curve



Product Specifications

Model	AT-70K-TLC-EU	AT-80K-TLC-EU	AT-90K-TLC-EU	AT-100K-TLC-EU	AT-110K-TLC-EU	AT-120K-TLC-EU
Input Data						
Max.DC input power	105kW	120kW	135kW	150kW	165kW	180kW
Max.DC input voltage	1100V					
Max.DC input current	40/40/20/40/40/20A	40/40/20/20/40/40/20/20A			45/20/45/20/45/20/45/20A	
MPPT voltage range	200-980V					
Recommended MPPT operating voltage	600V					
Starting voltage	180V					
Number Of MPPT	6	8				
Max. Number of strings per MPPT	2/2/1/2/2/1	2/2/1/1/2/2/1/1			3/1/3/1/3/1/3/1	
Output Data						
Rated output power	70kW	80kW	90kW	100kW	110kW	120kW
Max.output power	77kW	88kW	99kW	110kW	121kW	132kW
Max.output current	112A	128A	144A	160A	176A	191A
Rated grid voltage	400Vac					
Grid voltage range	310-480Vac					
Rated grid frequency	50Hz/60Hz					
Grid frequency range	45-55Hz/55-65Hz					
THD	< 3% (Under rated power)					
Power factor	> 0.99 (Under rated power) / Adjustable range: 0.8 leading-0.8 lagging					
DC current injection	< 0.5% (Under rated power)					
System Data						
Max. efficiency	98.60%	98.60%	98.60%	98.60%	98.70%	98.70%
European efficiency	98.10%	98.10%	98.10%	98.10%	98.10%	98.20%
Humidity range	0-100% non-condensing					
Cooling method	Intelligent forced air cooling					
Temperature range	-25°C to 60°C (Derating 45°C)					
Night-time power consumption	< 5W					
Max. working altitude	4000m(It needs to be derated when the altitude exceeds 2000m)					
Display	LED(optional: LCD)					
Communication interface	Wi-Fi(optional: RS485 or 4G)					
Protection						
DC reverse-polarity protection	YES					
Short circuit protection	YES					
Output overcurrent protection	YES					
Output overvoltage protection	YES					
Insulation resistance monitoring	YES					
Residual current detection	YES					
Surge protection	YES					
Grid monitoring	YES					
Islanding protection	YES					
Temperature protection	YES					
Integrated DC switch	YES					
Mechanical Data						
Dimensions (W*H*D)	1160*610*285mm					
Weight	84kg					
IP Rating	IP66					
Standard						
Safety standard/EMC standard	NB/T32004-2018, IEC62109, IEC61000, IS16169 & IS16221(BIS)					
Grid-connected standard	IEC61727, EN50549-1, VDE-4105, NRS-097-2-1, OVE-Richtlinie R25, UNE217001/2, Ordinance No.140					
Other standard	IEC61683, IEC62116, EN50530, IEC60068					
Optional Function						
Wi-Fi function	☐ YES ☐ NO					
Grid anti-backflow CT	☐ YES ☐ NO					

ALL-IN-ONE

ALL-IN-ONE

500W-1000WH-----	31-32
1000W-2000WH-----	33-34
5KW-10/15KWH-----	35-36
4.2KW-8KWH -----	37-38
6.2-12KW-10/16KWH -----	39-40
15KW-32KWH-----	41-42
7.2-12KW-16KWH(102V)-----	43-44

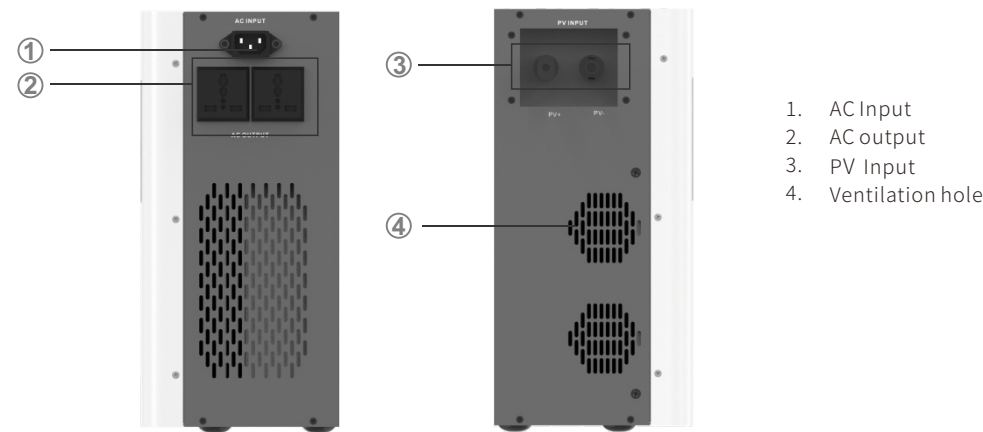
230V Series



500W-1000WH
ALL IN ONE

- Integrated design of inverter and BMS
- 6000/8000 times deep cycle charge and discharge.
- Support solar charging
- Wide MPPT Range 10Vdc-55Vdc
- Supports USB output & phone charging

Interface Description



Solar System Connection



Product Specifications

Model	MF-500W-1000WH-EU
AC Input/Output	
Rated AC Output Power	500W
Output Voltage Waveform	Pure Sine Wave
Output Voltage	185~230VAC, 50Hz
Nominal Input Voltage	230Vac
Low Loss Voltage	90Vac / 170Vac ± 7
Low Loss Return Voltage	100Vac / 180Vac ± 7
High Loss Voltage	280Vac ± 7
High Loss Return Voltage	270Vac ± 7
Max. AC Input Voltage	300Vac
Nominal Input Frequency	45~65Hz
AC Charging Power	500W Max.
PV Input	
Commonly used module power	150 to 550W, 1pcs
MPPT voltage range	10~50Vdc
Start-up voltage	10Vdc
Maximum input voltage	55Vdc
Maximum input power	400W Max.
DC Output	
DC output	12V2.5A DC Power JACK 5.5*2.1mm, 2pcs
USB Charger	5V1A, 1pcs
TYPE-C Charger	5V1A, 1pcs
Battery Cell	
Capacity	1004.8Wh
Nominal Battery Voltage	3.2Vdc
Typical Capacity	314Ah
Max. Charge Current	220A
Max. Discharge Current	220A
Operation Voltage Range	2.9Vdc~3.6Vdc
Others	
Cycle life	6000/8000 Cycles @ 90% DOD /25°C/0.5C
Placed	Indoors
Ingress Protection rating	IP21
Operating Temperature	-10°C~+50°C
Humidity	5%~95% Relative Humidity (Non-c condensing)
Altitude	≤2000 m
Product size	L325*W254*H153mm
Net Weight	11kg

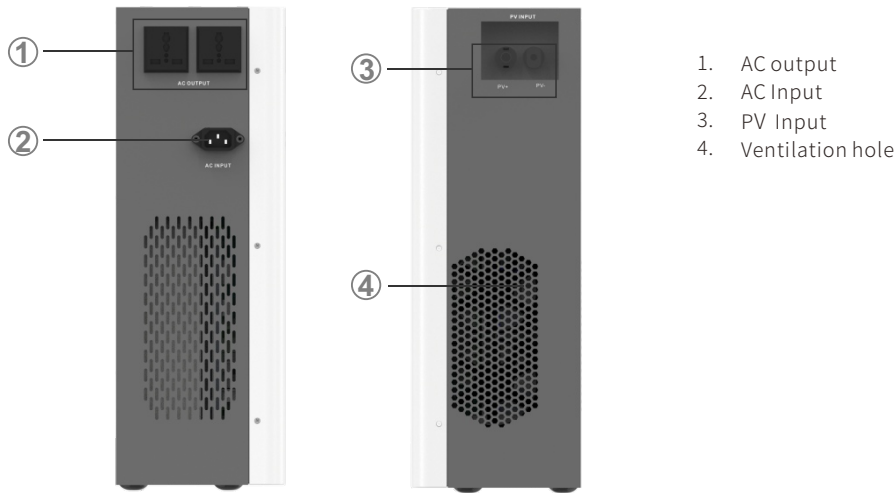
230V Series



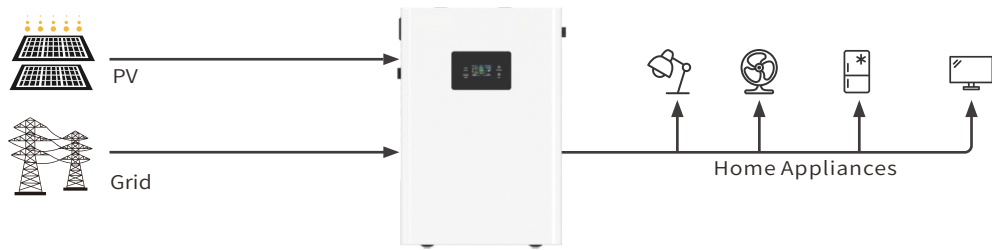
1000W-2000WH
ALL IN ONE

- Integrated design of inverter and BMS
- 6000/8000 times deep cycle charge and discharge.
- Support solar charging
- Wide MPPT Range 15Vdc-70Vdc
- Supports USB output & phone charging

Interface Description



Solar System Connection



Product Specifications

Model	MF-1000W-2000WH-EU
AC Input/Output	
Rated AC Output Power	1000W
Output Voltage Waveform	Pure Sine Wave
Output Voltage	185~230VAC, 50Hz
Nominal Input Voltage	230Vac
Low Loss Voltage	90Vac / 170Vac ± 7
Low Loss Return Voltage	100Vac / 180Vac ± 7
High Loss Voltage	280Vac ± 7
High Loss Return Voltage	270Vac ± 7
Max. AC Input Voltage	300Vac
Nominal Input Frequency	45~65Hz
AC Charging Power	1000W Max.
PV Input	
Commonly used module power	150 to 550W, 1pcs
MPPT voltage range	15~60Vdc
Start-up voltage	15Vdc
Maximum input voltage	70Vdc
Maximum input power	550W Max.
DC Output	
DC output	12V2.5A DC Power JACK 5.5*2.1mm, 2pcs
USB Charger	5V1A, 1pcs
TYPE-C Charger	5V1A, 1pcs
Battery Cell	
Capacity	2009.6Wh
Nominal Battery Voltage	6.4Vdc
Typical Capacity	314Ah *2
Max. Charge Current	220A
Max. Discharge Current	220A
Operation Voltage Range	5.8Vdc~7.2Vdc
Others	
Cycle life	6000/8000 Cycles @ 90% DOD /25°C/0.5C
Placed	Indoors
Ingress Protection rating	IP21
Operating Temperature	-10°C~+50°C
Humidity	5%~95% Relative Humidity (Non-c condensing)
Altitude	≤2000 m
Product size	L440*W310*H153mm
Net Weight	18kg

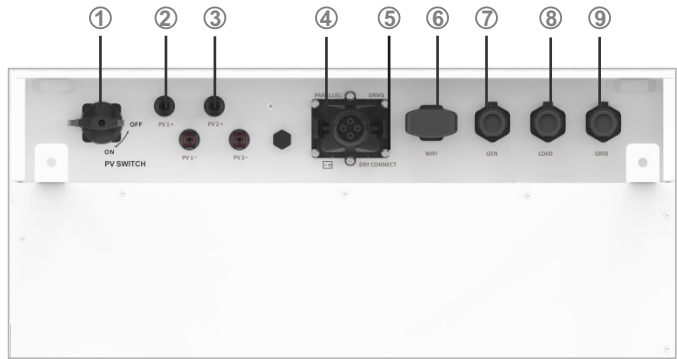
230V Series



5KW-10/15KWH
IP65 Stacked Energy Storage System

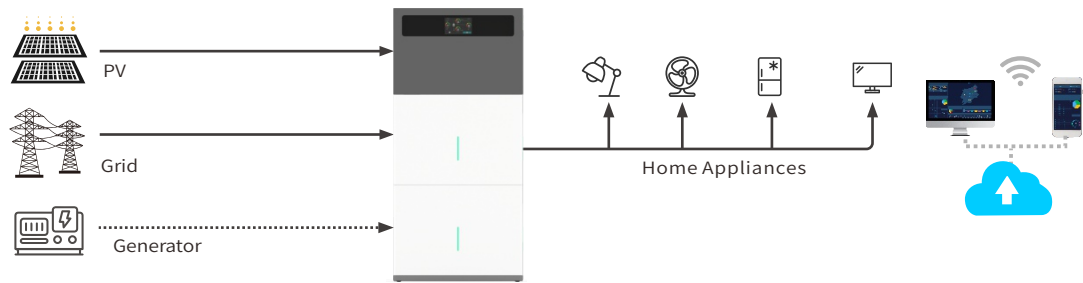
- **Less Waste, Max Storage**
High energy density and efficient energy utilization
- **Safe and Powerful Technology**
LFP (LiFePO4) battery cell, electrical isolation, external short circuit protection
- **Power for Your Home**
This system stores energy, detects outages, and automatically powers your home when the grid fails. Unlike generators, it provides quiet, maintenance-free backup—no fuel, no fumes. Pair it with solar panels to recharge with sunlight and keep your essentials running for days.

Interface Description



1. PV Switch
2. PV 1
3. PV 2
4. Parallel connection terminal
5. Dry contact control
6. Wi-Fi dongle interface
7. Generator
8. Load port
9. Grid interface

Solar System Connection



Product Specifications

Model	MF-5K-10KWH-EU		MF-5K-15KWH-EU
Battery Data			
Battery type	LiFePO4		
Battery nominal voltage (Vdc)	51.2		
Battery capacity (Wh)	5120X2	5120X3	
Number of Batteries	2	3	
Max. charging current/output current (A)	100		
Max. Rated discharge power (W)	5000		
Charging temperature(°C)	0~55		
Discharging temperature(°C)	-20~55		
Cycle life	6000/8000 Cycles @ 90% DOD/25°C/0.5C		
Output Data (Off-Grid)			
Rated Output Power	5000W		
Max. Peak Power	12,000VA		
Rated Output Voltage	230Vac, single phase		
Load Capacity of Motors	4HP		
Output Frequency Range	50Hz ± 0.3Hz/60Hz ± 0.3Hz		
Transfer time	≤5ms		
Output Data (On-Grid)			
Rated Output Power	5000W		
Max. Apparent Power	12000VA		
Rated. Output Current	24.2A		
Max. Output Current	26.6A		
THD	< 3%		
Power Factor	0.8 leading to 0.8 lagging		
Rated Voltage	230Vac, single phase		
Frequency	50Hz/60Hz		
PV Input			
Number of MPPT	2		
Max. Input Power	4500W+4500W		
Max. Input Current	18A+18A		
Max. Short-circuit Current	27A+27A		
Max. Open-circuit Voltage	500Vdc+500Vdc		
MPPT Operating Voltage Range	120~450Vdc		
Utility /Generator Input			
Input Voltage Range	90~280Vac		
Input Frequency Range	50/60Hz		
Bypass Overload Current	40A		
Efficiency			
MPPT Tracking Efficiency	99.9%		
Max. Efficiency	97.5%		
General Data			
Number of Parallel	1~6 units		
Operating Temperature Range	-20°C to 60°C (Derating 45°C)		
Humidity Range	0% ~ 100%		
Noise	<60dB		
Cooling method	heatsink & intelligent air cooling		
IP Rating	IP65		
Inverter Size(LxWxH) (MM)	600X200X400		
Package Size (LxWxH) (MM)	695X310X560		
Battery pack Size (LxWxH) (MM)	600X200X420		
Package Size (LxWxH) (MM)	680X280X510		
Inverter Net Weight/Gross Weight (kg)	33/35KG		
Battery pack Net Weight/Gross Weight (kg)	55/56.5KG		
Communication			
Internal Interface	Rs485 / CAN / USB / Dry contact control		
External Module (optional)	Wi-Fi / 4G		
Certification			
Inverter Certification	EN IEC 61000-6-:2019/EN IEC 61000-6-3:2021/AS 4777.2-2020/Amd 2-2024/IEC 60529 (IP) IEC 62109-1:2010/EN 62109-1:2010/IEC 62109-2:2011/EN 62109-2:2011/AS 60947.3:2023		
Battery pack certification	AS 60947.3/IEC 60529 (IP)		
Optional Function			
Parallel function	☐ YES	☐ NO	
Wi-Fi function	☐ YES	☐ NO	
Grid anti-backflow CT	☐ YES	☐ NO	

230V Series



4.2KW-8KWH

ALL IN ONE



Less Waste, Max Storage

Harness more of the sun's power—this energy storage system increases the self-consumption rate up to 90%, lowering your electricity bill and your carbon



Safe and Powerful Technology

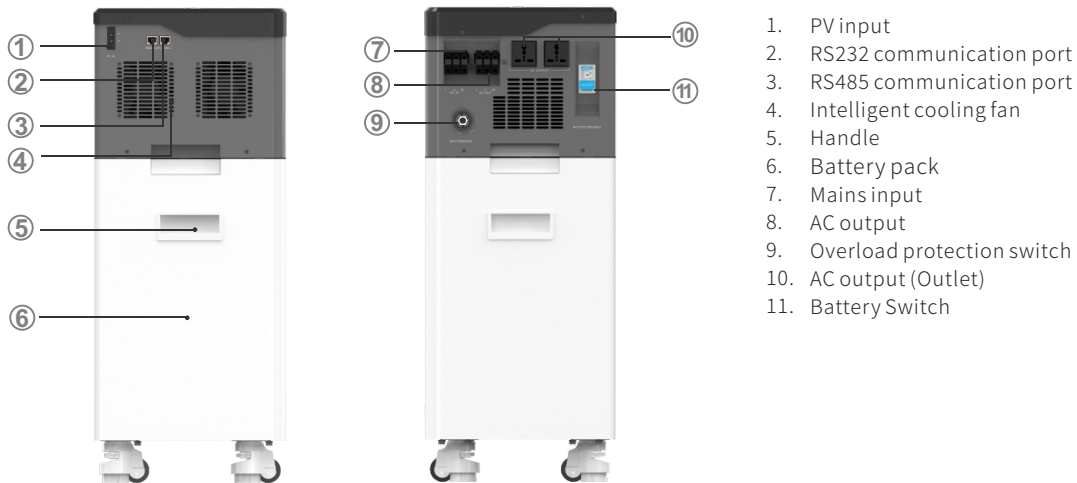
LFP (LiFePO4) batteries are safer, and an electrical isolation design protects against accidental short circuits



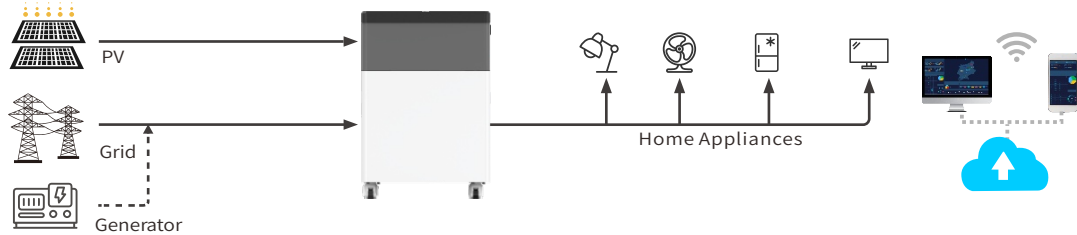
Power for Your Home

This system stores energy, detects outages, and automatically powers your home when the grid fails. Unlike generators, it provides quiet, maintenance-free backup—no fuel, no fumes. Pair it with solar panels to recharge with sunlight and keep your essentials running for days.

Interface Description



Solar System Connection



Product Specifications

Model	MF-4.2K-8KWH-EU
Battery Data	
Battery type	LiFePO4
Battery nominal voltage (Vdc)	51.2
Battery capacity (Wh)	7168
Max. charging current/output current (A)	80
Rated charging power/Rated discharge power (W)	4200
Charging temperature(°C)	0~55
Discharging temperature(°C)	-20~55
Cycle life	6000/8000 Cycles @ 90% DOD/25°C/0.5C
Input Data (PV)	
Max. input Power (W)	6000
Max. input voltage (Vdc)	500Vdc
MPPT Operating Voltage Range(Vdc)	120-450
Start voltage (Vdc)	120
Single MPPT input current (A)	20
Max. input current of single MPPT (A)	22
Number of MPPT	1
Number of Strings Per MPPTs	2
AC Output (Grid connected)	
Grid connected output nominal power (VA)	4200
Max. power output/input for grid connection (VA)	4200
Rated output voltage (Vac)	230
Output voltage range (Vac)	180-270
Rated operating frequency (Hz)	50/60(±0.2%)
Max. output/input current for grid connection (A)	16/18.3
Transfer time (ms)	10
Power factor	0.8 leading... 0.8 lagging
THD	<3%
AC Output (Off Grid)	
Off grid output nominal power (VA)	4200
Max. output power of off grid (VA)	4200
Max. off-grid output current (A)	18.3
Rated output voltage (Vac)	230(±2%)
Output voltage range (Vac)	180-270
Rated operating frequency (Hz)	50/60(±0.2%)
THD	<3%
Efficiency	
Max. Efficiency	>95%
Max. DC-AC transformation efficiency	>92%
MPPT Efficiency	99.9%
General Data	
Operating Temperature (°C)	-20°C to 60°C (Derating 45°C)
Relative Humidity	5%-95%
Noise(Normal)(dB)	≤60
Communication	RS232/RS485
Cooling method	FAN Cooling
IP Rating	IP21
Inverter Size (LxWxH) (MM)	520X240X304
Package Size (LxWxH) (MM)	610X340X360
Battery pack Size(LxWxH) (MM)	520X450X304
Package Size(LxWxH) (MM)	610X555X360
Inverter Net Weight/Gross Weight (kg)	14/16KG
Battery pack Net Weight/Gross Weight (kg)	62.6/64.5KG

230V Series



6.2/12KW-10/16KWH

ALL IN ONE



Less Waste, Max Storage

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Safe and Powerful Technology

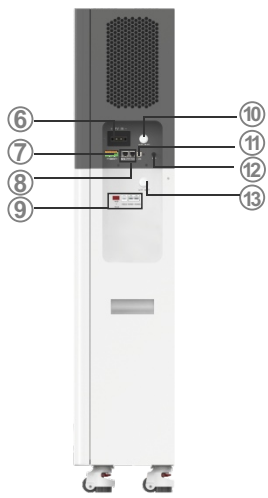
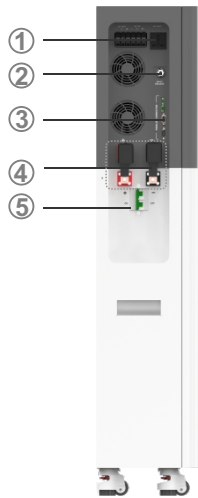
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Power for Your Home

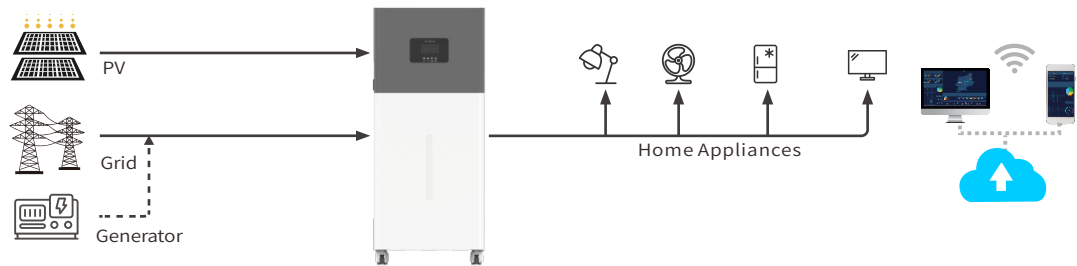
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Interface Description



- 1. AC OUT /AC IN
- 2. AC INPUT BREAKER
- 3. Parallel communication port
- 4. BAT+ BAT-
- 5. Battery breaker
- 6. PV input
- 7. Dry contact
- 8. RS485/BMS communication port
- 9. Battery communication port
- 10. ON/OFF Switch
- 11. USB communication port
- 12. Wi-Fi port
- 13. Battery ON/OFF Switch

Solar System Connection



Product Specifications

Model	MF-6.2K-10/16KWH-EU		MF-12K-16KWH-EU
Battery Data			
Battery type	LiFePO4		
Battery nominal voltage (Vdc)	51.2		
Battery capacity (Wh)	10240/16076		16076
Max. charging current/output current (A)	100/150		150
Rated charging power/Rated discharge power (W)	6200		12000
Charging temperature(°C)	0~55		
Discharging temperature(°C)	-20~55		
Cycle life	6000/8000 Cycles @ 90% DOD/25°C/0.5C		
AC mode			
Rated input voltage	230Vac ±5%		230Vac ±5%
Input voltage range	170~280VAC (UPS mode) ; 90~280VAC (APL mode)		
Frequency	50Hz/60Hz (Auto detection)		
Frequency Range	47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz);		
Overload/short circuit protection	Circuit breaker		
Efficiency	>93%		
Transfer time	≤5ms		
AC reverse protection	Yes		
Maximum bypass overload current	40A		60A
Inverter mode			
Output voltage waveform	Pure sine wave		
Rated output power (VA/W)	6200VA/6200W		12000VA/12000W
Power factor	1		
Rated output voltage (Vac)	230Vac		
Output voltage error	±5%		
Output frequency range (Hz)	50Hz±0.3Hz/60Hz±0.3Hz		
Maximum Efficiency	>92%		>94%
Peak power	12000VA		24000VA
Loaded motor capability	4HP		8HP
Output short circuit protection	Circuit breaker		
Bypass circuit breaker specification	40A		63A
Rated battery input voltage	51.2V (Minimum starting voltage 44V)		
Battery voltage range	40.0Vdc~60Vdc±0.6Vdc (Undervoltage alarm/shutdown voltage/overvoltage alarm /overvoltage recovery... settable on LCD screen)		
Power saving mode self-consumption	Load≤25W		Load≤50W
AC charging			
Battery type	Lead acid or lithium battery		
Maximum charge current(can be set)	80A		150A
Charge current error	±5Adc		
Charge voltage range	40 ~60Vdc		
Short circuit protection	Circuit breaker and blown fuse		
Circuit breaker specifications	40A		63A
PV charging			
MPPT way	1		2
Maximum PV open circuit voltage	500Vdc		
PV operating voltage range	120Vdc~500Vdc		120Vdc~500Vdc
MPPT voltage range	90Vdc~450Vdc		90Vdc~450Vdc
Battery voltage range	40-60Vdc		
Maximum PV input power	6500W		9000WX2
Maximum PV input current	27A		27A+27A
Charging short circuit protection	Blown fuse		
Wiring protection	Reverse polarity protection		
Hybrid Max. Charging Current(AC+PV)			
Max charging current(can be set)	0-100A		0-150A
General Data			
Operating temperature range	-20°C to 60°C (Derating 45°C)		
Storage temperature range	-25°C to 60°C		
Humidity range	5% to 95% (Conformal coating protection)		
IP Rating	IP21		
Cooling method	Forced air cooling, variable speed of fan		
Communication interface	USB/RS485(Wi-Fi/4G)/ Dry contact control		
Wi-Fi function	YES		
Product Size (LxWxH) (MM)	552X254X1210		
Packing Size (LxWxH) (MM)	635X335X1370		
Net Weight/Gross Weight (kg)	105/120	130/150	130/150
Optional Function			
Parallel function	☐ YES ☐ NO		
Grid anti-backflow CT	☐ YES ☐ NO		

230V Series



15KW-32KWH

ALL IN ONE



Less Waste, Max Storage

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Safe and Powerful Technology

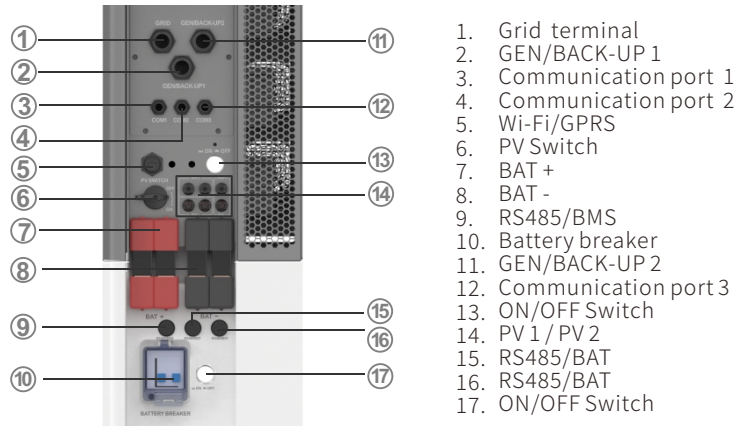
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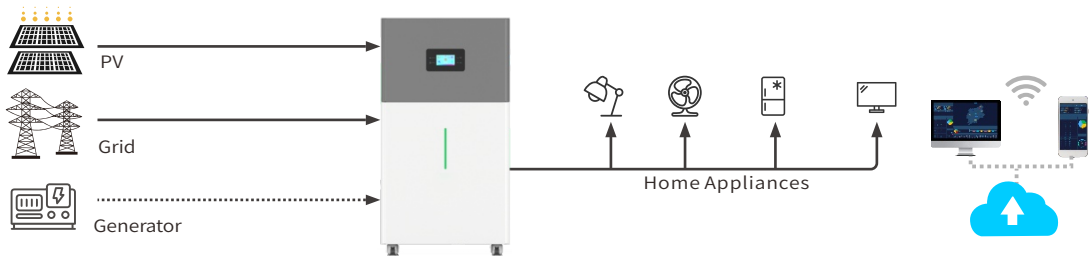
Power for Your Home

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Interface Description



Solar System Connection



Product Specifications

Model	MF-15K-32KWH-EU
Battery Data	
Battery type	LiFePO4
Battery nominal voltage (Vdc)	51.2
Battery capacity (Wh)	32150
Max. charging current/output current (A)	300
Rated charging power/Rated discharge power (W)	15000
Charging temperature(°C)	0~55
Discharging temperature(°C)	-20~55
Cycle life	6000/8000 Cycles @ 90% DOD/25°C/0.5C
PV Input Data	
Max. DC Input Power(KW)	30KW
Max. DC Input Voltage(V)	1000
MPPT Voltage Range (V)	200~800V
Start-up Voltage (V)	150V
Max. Input Current(A)	36A/18A
Number of MPPT Trackers	2
AC Output Data(Grid-tied)	
Max. Apparent Output Power(VA)	16500
Rated Output Voltage(V)	380/400V
Rated Output Frequency (Hz)	50/60Hz
Maximum Output Current (A)	25A
Output Power Factor	1(0.8 leading~0.8lagging)
AC Output Data (Off-grid)	
Rated Output Power(W)	
Peak Output Power(W)	15000
Rated Output Voltage (V)	>200%,15sec
Rated Output Frequency (Hz)	380/400V
Rated Output Current(A)	50/60Hz
Efficiency	
Max.Efficiency(PV)	
Max.Efficiency(Battery)	98%
European Efficiency	94.5%
Protection	
PV Perverse Polarity Protection	Integrated
PV Insulation Resistance Detection	Integrated
Residual Current Detection	Integrated
Output Overcurrent Protection	Integrated
Output Short-circuit Protection	Integrated
Output Overvoltage Protection	Integrated
General Data	
Operating Temperature(°C)	-20°C to 60°C (Derating 45°C)
Storage Temperature(°C)	-30 to 65°C
Relative Humidity	0~95%
Operating Altitude (m)	≤4000m(derating above 2000m)
Cooling method	Intelligent Forced Air Cooling
Noise(dB)	<55dB
IP Rating	IP66
Topology	High-frequency Isolation(Battery Side)
Product Size (LxWxH) (MM)	760X304X1592
Packing Size (LxWxH) (MM)	900X553X1720
Net Weight/Gross Weight (kg)	300/330
Optional Function	
Parallel function	☐ YES ☐ NO
Wi-Fi function	☐ YES ☐ NO
Grid anti-backflow CT	☐ YES ☐ NO

120V Series

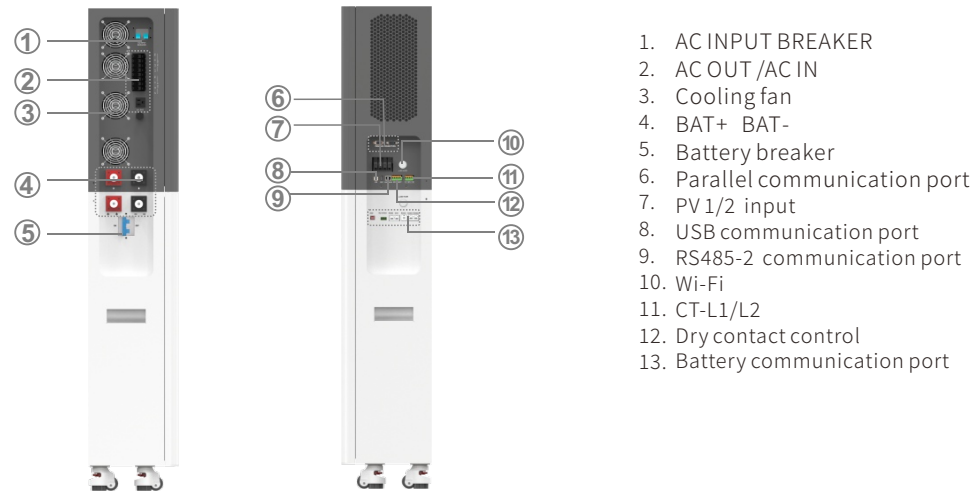


7.2/12KW-16KWH

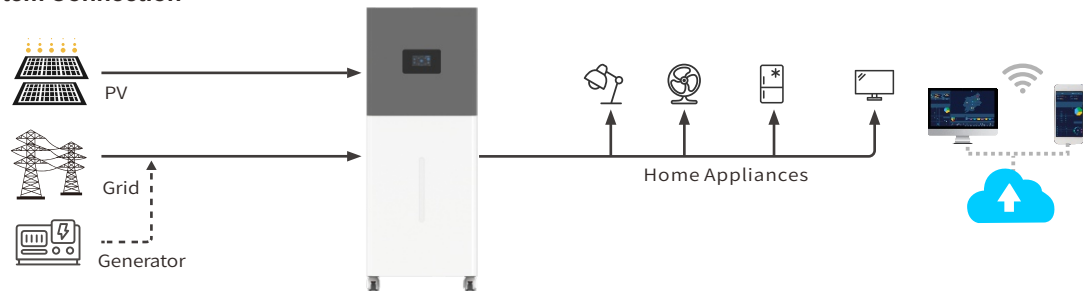
ALL IN ONE

-  **Less Waste, Max Storage**
High energy density and efficient energy utilization
-  **Safe and Powerful Technology**
LFP (LiFePO4) battery cell, electrical isolation, external short circuit protection
-  **Power for Your Home**
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Interface Description



Solar System Connection



Product Specifications

Model	MF-7.2K-16KWH-US		MF-12K-16KWH-US	
Battery Data				
Battery type			LiFePO4	
Battery nominal voltage (Vdc)			51.2	
Battery capacity (Wh)			16076	
Max. charging current/output current (A)			150	
Rated charging power/Rated discharge power (W)	7200		12000	
Charging temperature(°C)			0~55	
Discharging temperature(°C)			-20~55	
Cycle life			6000/8000 Cycles @ 90% DOD/25°C/0.5C	
AC Mode				
Rated input voltage			120(L1/N, L2/N)/240Vac(L1/L2)	
Input phase voltage range			(85Vac~140Vac)±2%	
Frequency			50Hz/60Hz (Automatic detection)	
Frequency Range			47±0.3Hz~55±0.3Hz (50Hz); 57±0.3Hz~65±0.3Hz (60Hz);	
Overload/short circuit protection			Circuit breaker	
Efficiency			>95%	
Transfer time			≤5ms	
AC reverse protection			Yes	
Maximum bypass phase overload current			2 pole, 63A/63A(L1/L2)	
Inverter Mode				
Output voltage waveform			Pure sine wave	
Rated output power (VA)	Split phase: 7200 Single phase: 6000 (limited by the wiring terminals)		Split phase: 12000 Single phase: 7200 (limited by the wiring terminals)	
Rated output power (W)	Split phase: 7200 Single phase: 6000		Split phase: 12000 Single phase: 7200 (limited by the wiring terminals)	
Power factor			1	
Rated output voltage (Vac)			120Vac (L1/N, L2/N)/240Vac(L1/L2)	
Output voltage error			±5%	
Output frequency range (Hz)			50Hz±0.3Hz/60Hz±0.3Hz	
Maximum Efficiency			>91%	
Peak power	Split phase: 10800VA Single phase: 10800VA		Split phase: 18000VA Single phase: 18000VA	
Loaded motor capability	4HP		6HP	
Output short circuit protection			Can protect, after continuous short circuit for 1 second	
Bypass circuit breaker phase specification			2 pole, 63A/63A(L1/L2)	
Rated battery input voltage			51.2V (Minimum starting voltage 44V)	
Battery voltage range	40.0Vdc~58.6Vdc±0.6Vdc		(Undervoltage alarm/shutdown voltage/overvoltage alarm /overvoltage recovery... setttable on LCD screen)	
Power saving mode self-consumption			Load≤25W enter power-saving mode	
AC Charging				
Battery type			Lead acid or lithium battery	
Maximum charge current(can be set)	120A		160A	
Charge current error			±5Adc	
Charge voltage range			40 ~58.6Vdc	
Short circuit protection			Circuit breaker and blown fuse	
Circuit breaker phase specifications			2 pole, 63A/63A(L1/L2)	
PV Charging				
Number of MPPT groups			2	
Maximum PV open circuit voltage			500Vdc	
PV operating voltage range	90Vdc~485Vdc		90Vdc~500Vdc	
MPPT voltage range			90Vdc~450Vdc	
Battery voltage range			40-58.6Vdc	
Maximum PV input power	4500W+4500W		9000W+9000W	
Maximum PV input current	16A+16A		27A+27A	
Charging short circuit protection			Blown fuse	
Hybrid Max. Charging Current(AC+PV)				
Max charger current(can be set)	0-150A		0-220A	
General Data				
Operating temperature range			-20°C to 60°C (Derating 45°C)	
Storage temperature range			-25°C to 60°C	
Humidity range			5% to 95% (Conformal coating protection)	
Noise			≤60dB	
Cooling method			Forced air cooling, variable speed of fan	
Communication interface			USB/RS485(Wi-Fi/4G)/ Dry contact control	
IP Rating			IP21	
Optional Function				
Parallel function			☐ YES	☐ NO
Wi-Fi function			☐ YES	☐ NO
Grid anti-backflow CT			☐ YES	☐ NO

BATTERY

Wall mounted & Floor Standing

5/10KWH -----47

Moveable & Floor Standing

16KWH -----48

Wall mounted & Floor Standing



5/10KWH

- More Usable Energy**
90% depth of discharge
- High Efficiency**
Full-time 1C discharge
- High Expansibility**
Max 15 batteries in parallel. Scalable from 5KWh to 150KWh
- Flexible Installation**
Wall mounted or floor stand
- Safe&Reliable**
Lithium iron phosphate cell Cobalt free
- Perfect Compatibility**
Compatible with mainstream inverter brand

Moveable & Floor Standing



16KWH

- More Usable Energy**
90% depth of discharge
- High Energy**
Single cell with 314Ah
- High Expansibility**
Max 15 batteries in parallel. Scalable from 16KWh to 240KWh
- Easy Installation**
Floor stand with wheels
- Perfect Compatibility**
Compatible with major inverter brand
- Smart Display Screen**
Touchscreen-control.Real-time battery status display

Product Specifications

Model	MZ-5KWH	MZ-10KWH
Performance		
Nominal Voltage	51.2 Vdc	51.2 Vdc
Nominal Capacity	100 Ah	200 Ah
Battery Energy	5120Wh	10240Wh
Rated charging/discharging current	50 A	50 A
Max charging/discharging current	100 A	100 A
Charging voltage	58.4V	58.4V
Communication		
Display	LED indicator+ LCD Display	LED indicator+ LCD Display
Communication	RS232、RS485、CAN	RS232、RS485、CAN
General Specification		
Size/Package Size(LxWxH) (MM)	433.5x168.5x580/515x376x673mm	498x262.7x660/586x481x772mm
Weight (Kg)	43Kg±3kg	80Kg±3kg
Installation	Floor Stand or Wall Mount	Floor Stand or Wall Mount
Operating temperature range	-20°C to 55°C (Derating 45°C)	-20°C to 55°C (Derating 45°C)
Operating/Storage Humidity	≤95%RH	≤95%RH
Max Operating Altitude	≤2000m	≤2000m
IP Rating	IP21	IP21
Cell Technology	LiFePO4, Lithium Iron Phosphate	LiFePO4, Lithium Iron Phosphate
Cycle life	6000/8000 Cycles @ 90% DOD/25°C/0.5C	6000/8000 Cycles @ 90% DOD/25°C/0.5C
Scalability	Max 15 batteries in parallel	Max 15 batteries in parallel
Warranty	5 years	5 years
Standard Compliance (More Available Upon Request)		
Certificates	CE, UN38.3	
Ordering and Deliverable Part		
Product ordering part	Battery (Cable /Wall Mount Included)	

Product Specifications

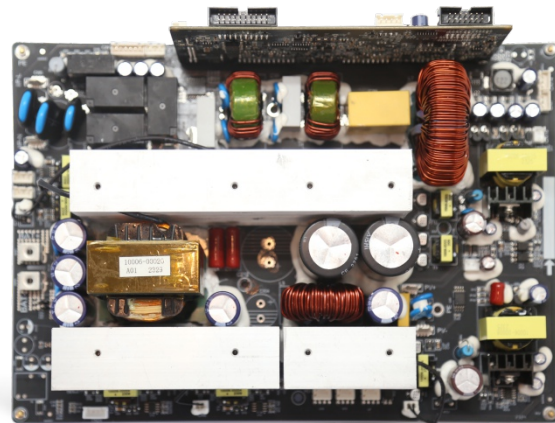
Model	MZ-16KWH
Performance	
Nominal Voltage	51.2 Vdc
Nominal Capacity	314Ah
Battery Energy	16076Wh
Rated charging/discharging current	157 A
Max charging/discharging current	200 A
Charging voltage	58.4V
Communication	
Display	LCD Display
Communication	RS232、RS485、CAN
General Specification	
Size/Package Size(LxWxH) (MM)	430x261x970/520x450x1070mm
Weight (kg)	114Kg±5%kg
Installation	Floor Stand
Operating temperature range	-20°C to 55°C (Derating 45°C)
Operating/Storage Humidity	≤95%RH
Max Operating Altitude	≤2000m
IP Rating	IP21
Cell Technology	LiFePO4, Lithium Iron Phosphate
Cycle life	6000/8000 Cycles @ 90% DOD/25°C/0.5C
Scalability	Max 15 batteries in parallel
Warranty	5 years
Standard Compliance (More Available Upon Request)	
Certificates	CE, UN38.3
Ordering and Deliverable Part	
Product ordering part	Battery (Cable /Wall Mount Included)

PCBA SKD

Inverter PCBA

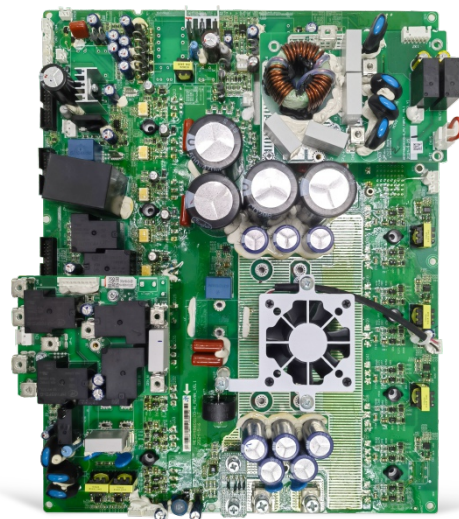
On & Off Grid Inverter PCBA -----	51
Hybrid Inverter PCBA -----	51
On Grid Inverter PCBA -----	52

PCBA SKD



On & Off Grid Inverter PCBA

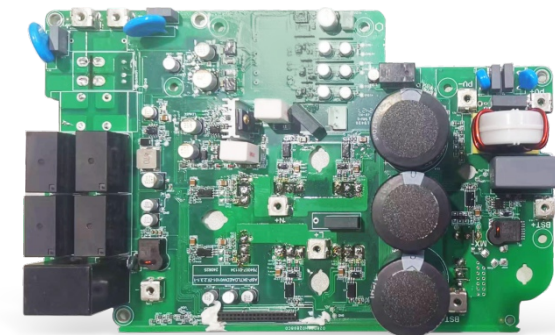
- ✓ 220/120V
- ✓ IP21/IP54
- ✓ 1.2-12KW



Hybrid Inverter PCBA

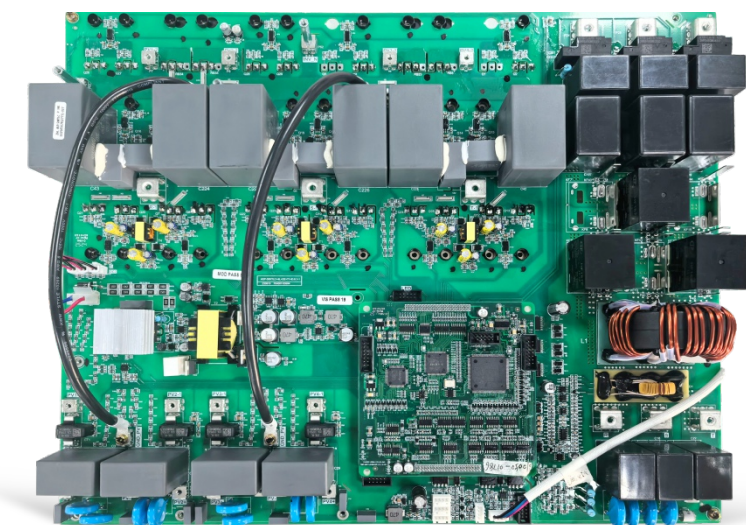
- ✓ 220/380V
- ✓ IP66
- ✓ 6-20KW

PCBA SKD



On Grid Inverter PCBA

- ✓ 220V
- ✓ IP66
- ✓ 2-10KW



On Grid Inverter PCBA

- ✓ 380V/500V/800V
- ✓ IP66
- ✓ 36-120KW